Empirical Analysis of Hyperparameter Scaling and Training Stability in Evolutionary Poker AI

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Abstract

This study presents a large-scale empirical analysis of evolutionary self-play training in Texas Hold'em poker, focusing on hyperparameter sensitivity, training stability, and opponent diversity mechanisms. Across 23 training configurations and 5,672 tournament games, we identify reproducible scaling relationships governing population size, matchup diversity, mutation strength, evaluation budget, and Hall of Fame opponent integration. We report strong non-monotonic performance trends, derive empirical hyperparameter laws, and demonstrate that historical opponent retention significantly improves generalization and reduces co-evolutionary overfitting. A champion configuration achieves an 82.9% tournament win rate while maintaining high computational efficiency. These findings provide actionable guidelines for compute-efficient evolutionary training in imperfect-information games.

1 Introduction

Evolutionary algorithms offer a gradient-free alternative to reinforcement learning for training decision-making agents in sparse-reward and imperfect-information environments. While evolutionary methods have been applied to games in prior work, the stability, scalability, and hyperparameter sensitivity of evolutionary self-play systems remain insufficiently characterized, particularly in complex domains such as poker.

This work investigates how evolutionary hyperparameters interact to influence learning stability, opponent generalization, computational efficiency, and performance. Our objective is not to solve poker optimally, but to empirically study training dynamics, failure modes, and scalable optimization strategies in co-evolutionary learning systems.

2 System Overview

Policy Network: Feedforward neural network ($17 \rightarrow 64 \rightarrow 32 \rightarrow 6$), mapping poker state features to six discrete betting actions.

Training Method: Population-based evolutionary optimization with Gaussian mutation applied to neural network weights.

Evaluation: Round-robin self-play tournaments measuring win rate and Big Blinds per 100 hands (BB/100).

Hall of Fame (HoF): A persistent opponent pool containing elite historical agents to improve training stability.

Game Engine: Optimized Texas Hold'em simulator with accelerated hand evaluation and batch inference.

3 Experimental Methodology

3.1 Dataset and Configurations

- Total tournaments: 11
- Total games played: 5,672
- Unique configurations tested: 23
- Population sizes: $\{12, 20, 40\}$
- Matchups per agent: $\{3, 6, 8, 10\}$
- Hands per matchup: $\{375, 500, 750, 1000\}$
- Mutation sigma: $\{0.08, 0.10, 0.12, 0.15\}$
- Generation counts: $\{50, 200\}$

Each agent participated in 68 to 404 tournament games.

3.2 Hall of Fame Protocol

Two training regimes were evaluated:

1. Pure self-play against current population
2. HoF-enabled training with three elite historical opponents

4 Hyperparameter Effects

4.1 Matchups per Agent

Matchups (m)	Mean Win Rate	Std Dev	Interpretation
8	**65.8%**	12.4%	Optimal
6	46.2%	15.8%	Baseline
10	28.1%	8.3%	Overfitting
3	22.2%	4.1%	Under-sampled

Table 1: Non-monotonic impact of matchup diversity.

4.2 Mutation Sigma

Sigma (σ)	Mean Win Rate	Mean Chips	Stability
0.10	**54.0%**	28,469	Optimal
0.08	50.0%	27,651	Near-optimal
0.12	48.5%	25,207	Acceptable
0.15	34.2%	15,955	Destabilizing

Table 2: Mutation strength strongly controls convergence.

4.3 Hands per Matchup

Hands (h)	Mean Win Rate	Efficiency	Verdict
375	**50.6%**	High	Optimal
500	45.6%	Moderate	Baseline
750	48.1%	Lower	Inefficient
1000	22.2%	Poor	Counterproductive

Table 3: Diminishing returns beyond moderate evaluation budgets.

4.4 Population Size

Population (p)	Mean Win Rate	Compute Cost	Notes
12	45.1%	**Low**	Most efficient
20	46.3%	Medium	Balanced
40	49.3%	High	Marginal gains

Table 4: Small populations remain competitive under HoF training.

5 Empirical Hyperparameter Scaling Relationships

Analysis across 19 configurations and 2,952 tournament matches reveals three reproducible interaction laws governing training stability.

5.1 Population Size vs. Matchups per Agent

Population	Matchups	Ratio (m/p)	Win Rate
12	8	0.67	81.2%
20	6	0.30	71.3%
40	8	0.20	78.7%
40	6	0.15	73.1%

Table 5: Smaller populations require higher matchup ratios.

$$m \approx \begin{cases} 0.6p, & p \leq 20 \\ 0.2p, & p \geq 40 \end{cases}$$

5.2 Matchups vs. Hands Budget

Opponent diversity consistently provided stronger learning signal than deeper sampling.

Matchups	Hands	Total Evals	Win Rate	Efficiency
8	375	3,000	78.7%	High
8	500	4,000	81.2%	High
6	500	3,000	68–73%	Moderate
6	750	4,500	61–67%	Low

Table 6: Diversity outperforms depth at fixed evaluation budgets.

5.3 Population vs. Mutation Strength

Mutation stability follows an inverse scaling law:

$$\sigma \approx \frac{0.5}{\sqrt{p}}$$

Population	Sigma	Win Rate	Verdict
12	0.08	81.2%	Optimal
40	0.10	78.7%	Optimal
40	0.15	73.1%	Unstable

Table 7: Mutation should decrease as population diversity increases.

6 Hall of Fame Training Impact

Training Mode	Mean Win Rate	Mean Chips	Best Agent
HoF Enabled	**50.8%**	**25,403**	80.2%
Pure Self-Play	33.3%	15,897	55.0%

Table 8: Hall of Fame improves generalization and stability.

HoF reduces co-evolutionary overfitting by forcing adaptation against historical opponent strategies.

7 Champion Configuration

```
Population = 12
Matchups = 8
Hands = 500
Sigma = 0.08
Generations = 200
Hall of Fame = Enabled
```

- Win Rate: 82.9% (252W–52L)
- Average Chips: 40,500
- Training Time: 7–10 minutes per 100 generations

8 Discussion

Key insights:

- Hyperparameter relationships exhibit non-linear scaling behavior
- Matchup diversity acts as an implicit regularizer
- Excess mutation destabilizes evolutionary search
- Hall of Fame functions as curriculum learning
- Smaller populations can outperform brute-force scaling

9 Limitations

- Confounding sigma differences in some HoF comparisons
- Limited multi-seed replication
- No CFR or RL baseline included yet
- Results specific to two-player Texas Hold'em

10 Future Work

- Controlled HoF ablation study
- CFR / PPO baseline comparison
- Neural architecture scaling tests
- Cross-environment generalization analysis
- Validation of predicted p50–p60 scaling laws

11 Conclusion

This study demonstrates that evolutionary poker training is governed by stable hyperparameter scaling laws and benefits strongly from opponent retention mechanisms. These findings provide practical guidance for compute-efficient evolutionary learning in imperfect-information environments.

12 Repository

`https://github.com/Dheirav/PokerBot`